What Has TabPFN's Prior Learned?

Visualizing the Inductive Bias of a Tabular Foundation Model

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1 Project Title and Team Information

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2 Research Paper and Critical Analysis

Background. TabPFN is a transformer pretrained only on synthetic tables drawn from a prior of random structural causal models [5, 6]. As a prior-data fitted network [10], it reads a training set as context and predicts in one forward pass, so its inductive bias is learned, not designed, and must be measured.

Key paper. McCarter [9], *What Exactly Has TabPFN Learned to Do?* (ICLR 2024 Blogposts Track; 2025 revision with a TabPFN v2 appendix).

Main ideas and methods. The paper treats a fitted TabPFN as a black-box function $f_{\mathcal{D},\theta} : x \mapsto y$ and plots it on small, hand-built training sets. In 1D it found non-monotonic probabilities and an off-center boundary that ensembling smoothed, odd effects of duplicated samples, and no periodic interpolation. In 2D with one point per class, a 32-member ensemble produced regions resembling a Voronoi diagram, i.e. 1-nearest-neighbor behavior. The v2 appendix reports "bumpy" probability curves and approximate learning of parity.

Contributions. A probing method for pretrained in-context learners, evidence of "brilliant or baffling" behavior, and the case for matching PFN priors to problem classes.

Critical analysis. The probing idea is powerful, but the evidence is qualitative: one-off configurations, no metrics or repeated seeds, and no classic baseline on the same points. It covers only v1 and v2, while TabPFN-3.5 is about 30 times larger than v2, encodes cells with learned Fourier features and empirical-CDF ranks, and borrows from the TabICLv2 prior [12]. It never asks whether the bias *changes with the data*, and static figures cannot show a boundary evolving as context is added.

What we build on and extend. We adopt the probing method and turn its observations into testable hypotheses, then extend it four ways: *quantify* probes with metrics over seeds, *compare* against seven classic models, *re-test* on TabPFN-3.5, and *test adaptation* on two contrasting real datasets. Related work motivates each: neural networks favor smooth, rotation-invariant functions while trees win on medium tabular data [3]; TabPFN v2 adapts like a Gaussian process but is conservative on fast-varying functions [19]; in-context learners can form complex boundaries [2]; TabPFN v2 predicts via an attention-weighted vote over context labels [1]; and competing models inject tree-like priors [13, 14, 20], suggesting the prior shapes the boundary.

3 Proposed Project and Innovation

Research questions. RQ1: How do TabPFN's boundaries and uncertainty differ, measurably, from classic models on identical data? **RQ2:** Does TabPFN adapt its bias, behaving smooth on smooth data and step-like on rule-like data? **RQ3:** Did TabPFN-3.5's changes in scale, encoding and prior alter its bias relative to v2?

Connecting visualization and machine learning. Visualization is both our *measurement instrument*, since every metric is computed from the same dense prediction grid we draw, and our

explanation, since an interactive page lets readers add context points and watch an in-context learner update, which static figures cannot show.

	Hypothesis	Motivated by
H1	TabPFN's surfaces are smooth ramps: step score S (Eq. 1) near smooth models'	SCM prior [5]
H2	TabPFN under-fits high-frequency targets (tight spirals, fine checkerboards)	[19]
H3	TabPFN is humbler than trees far from data, though less than a GP at tiny n	[7]
H4	TabPFN resembles smooth models on smooth data and trees on rule-like data	mixed prior [6]
H5	3.5 is more step-like and more rotation-sensitive than v2	[12, 14]
H6	3.5 captures periodic stripes that v2 misses	[9, 12]

Approach and difference from existing work. Boundary plots have compared architectures and seeds [17] and reached high dimensions via projections [15]; we apply them to in-context learners, pair each picture with a metric, and add real-data interventions (Table 1).

	McCarter [9]	This project
Evidence	Qualitative, single configurations	Metrics (J, S, far-field, ECE), 3 seeds
2D baselines	None (Voronoi reference)	Seven classic models, same points
Model versions	v1; v2 in appendix	v2 and TabPFN-3.5 side by side
Data dependence	Not studied	Smooth vs. rule-like data; rotation, noise
Presentation	Static figures	Interactive explainer (Section 5)

Table 1: How the project builds on and differs from the key paper.

Expected contributions. (1) the first boundary-level characterization of TabPFN-3.5 next to v2 that we know of; (2) a direct test of bias adaptation on real data; (3) shape metrics that separate model families, released with code; and (4) an interactive explainer.

4 Project Plan and Evaluation

Synthetic data (RQ1, RQ3). Generated in code with scikit-learn [11] and NumPy: moons, circles, XOR, spirals at three tightness levels, periodic stripes, and 45°-rotated and 10–20% label-noise variants, with $n \in \{10, 30, 60, 200\}$ training points and three seeds.

Real data (RQ2). We test adaptation on two public OpenML datasets [18] chosen because they favor *opposite* biases (Table 2). Banknote Authentication [8] has four continuous wavelet statistics of banknote images, a setting where smooth models do well. Electricity is the version curated for the tree-favoring benchmark of Grinsztajn et al. [3], whose class labels mark whether the New South Wales price rose or fell relative to its 24-hour moving average. Published reference plots exist: Grinsztajn et al. [3] (Fig. 20) drew random-forest and MLP boundaries on its two most important features, and den Breejen et al. [2] (Fig. 1) did so for TabPFN using a different OpenML version (ID 44156).

Dataset	OpenML ID	Rows	Features	Class sizes	Role
Banknote Authentication	1462	1,372	4 numeric	762 / 610	smooth
Electricity (benchmark version)	44120	38,474	7 numeric	19,237 / 19,237	rule-like

Table 2: Real datasets as listed on OpenML; neither has missing values. Electricity's features are `date`, `period`, `nswprice`, `nswdemand`, `vicprice`, `vicdemand` and `transfer`.

Acquisition. Both are downloaded with `sklearn.datasets.fetch_openml(data_id=1462)` and `fetch_openml(data_id=44120)`. A data ID pins one fixed version; we cache each table as Parquet

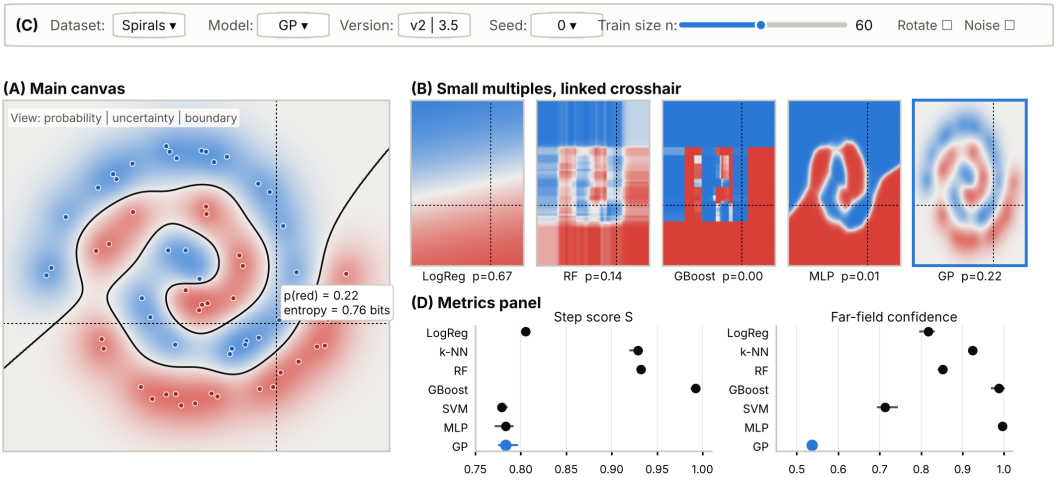


Figure 1: Pilot grids in the planned interface. (A) Canvas: GP probability of red on spirals (gray = uncertain, black = 0.5 boundary). (B) Five classic models on the same 60 points with a linked crosshair; trees give blocky regions. (C) Controls. (D) Metrics from Table 3.

with a checksum, and the ARFF files on the OpenML file server are the fallback. *Preprocessing.* Labels are mapped to $\{0, 1\}$. Neither dataset has missing values or categorical features. Features are standardized with training-split statistics only. We keep all seven Electricity features, as in the benchmark. *Sampling.* For each of three seeds, we draw a stratified training set of $n \in \{100, 500, 1000\}$ rows and a disjoint stratified test set of 300 rows, with the same sizes for both datasets so that size cannot explain differences. *Bias fingerprint.* First we compute $\Delta = a_{\text{tree}} - a_{\text{smooth}}$, the mean accuracy of RF and gradient boosting minus that of SVM, MLP and GP, with log loss as a second check. We expect $\Delta \leq 0$ on Banknote and $\Delta > 0$ on Electricity. If Banknote's accuracy saturates near 100%, we decide on log loss and the $n = 100$ setting. *Four adaptation tests*, all with identical inputs for every model: (1) *resemblance* on all features, $D(f, g) = \frac{1}{300} \sum_i |f(x_i) - g(x_i)|$, the mean absolute gap between TabPFN's and model g's probabilities on the test set; H4 predicts TabPFN's nearest model is smooth on Banknote and tree-based on Electricity; (2) *2D views*: all models retrained on the two features ranked highest by random-forest permutation importance on the training split, plotted like Figure 1B over the 1st–99th percentile range, and compared with the published Electricity plots; (3) *rotation*: the accuracy drop after multiplying the standardized features by a random orthogonal matrix (`scipy.stats.ortho_group`, one per seed and shared by all models), which removes axis-aligned structure [3]; and (4) *uninformative features*: the accuracy drop after appending as many $\mathcal{N}(0, 1)$ noise columns as there are real features. TabPFN adapts if its sensitivities track the trees' on Electricity and the smooth models' on Banknote.

Implementation. TabPFN v2 and 3.5 run through the official `tabpfn` package (default settings, plus 1, 4 and 8 ensemble members to test McCarter's ensembling effect), alongside logistic regression, k-NN, random forest, gradient boosting, RBF SVM, MLP and a Gaussian process with fixed, near-default settings. Synthetic runs predict on a 200×200 grid extending past the data; all predictions and metrics are saved once as JSON for the analyses and the interface. Compute: a few GPU-hours on free Colab; NYU HPC as backup.

Metrics. Accuracy, log loss and expected calibration error (ECE) [4]; seed stability [17]; *far-field confidence*, the mean maximum class probability outside the convex hull of the training points on a grid three times the data range; jaggedness J, adapted from den Breejen et al. [2]; and our step score S. With $\Delta^x_{ij} = p_{i,j+1} - p_{ij}$, $\Delta^y_{ij} = p_{i+1,j} - p_{ij}$, $|G|$ grid cells and C the cells where the prediction

changes,

$$J = \frac{1}{|G|}\sum_{i,j}\left(|\Delta^x_{ij}| + |\Delta^y_{ij}|\right), \qquad S = \frac{\sum_C \sqrt{(\Delta^x_{ij})^2 + (\Delta^y_{ij})^2}}{\sum_C \left(|\Delta^x_{ij}| + |\Delta^y_{ij}|\right)} \in \left[\tfrac{1}{\sqrt{2}}, 1\right]. \tag{1}$$

Step-like surfaces change along one grid axis at a time ($S \approx 1$); smooth ramps change along both (lower S).

Pilot results. Seven classic models, four synthetic datasets, 60 training and 60 test points, three seeds (Figure 1B, Table 3): J alone does not separate families, since gradient boosting's square circle scores *lower* than the MLP's round one (0.82 vs. 0.95). S does, with trees and k-NN at 0.93–0.99 and smooth models at 0.78–0.81 (seed s.d. ≤ 0.011), so H1 uses S. Only the GP becomes uncertain far from data (0.54 vs. 0.71–1.00), yet it has the highest ECE (0.166), so calibration and far-field humility are evaluated separately.

	Accuracy (%)	J circles	J spirals	S	Far-field conf.	ECE
Logistic regression	62.9 ± 2.6	0.28	0.48	0.81	0.82	0.144
k-nearest neighbors	88.5 ± 1.0	1.40	1.73	0.93	0.92	0.067
Random forest	89.7 ± 0.2	1.49	2.26	0.93	0.85	0.117
Gradient boosting	88.9 ± 1.5	0.82	2.14	0.99	0.99	0.112
SVM (RBF)	88.7 ± 0.4	0.94	0.96	0.78	0.71	0.119
Neural net (MLP)	91.7 ± 1.5	0.95	1.78	0.78	1.00	0.085
Gaussian process	92.6 ± 1.3	1.02	1.42	0.78	0.54	0.166

Table 3: Pilot baselines (means over three seeds; accuracy, S, far-field and ECE averaged over four datasets; ± = seed s.d.).

Evaluation. *(1) Hypotheses:* each of H1–H6 is decided by one pre-specified metric comparison over three seeds; differences within the seed spread count as null, and nulls are reported. *(2) Replication:* we first rerun McCarter's 1D and 2D probes on v2 to confirm our pipeline reproduces his findings. *(3) Interface effectiveness:* 6–8 classmates match unlabeled boundary plots to model families and locate overconfident regions before and after using the interface; we report accuracy, completion time and think-aloud notes. *(4) Validity:* 2D probes may not reflect higher dimensions, so RQ2 uses all real features; grid settings are fixed and rechecked at a second resolution.

Roles. Kiet Nguyen leads the experiments, TabPFN runs and metrics; Nhan Doan leads the datasets, baselines and the interface; both run the study and write the report. If time is short we drop, in order: the ensemble-size ablation, the noise-column test, and H6.

5 Visualization Interface

A single D3.js page on GitHub Pages, modeled on TensorFlow Playground [16] but showing what an in-context learner predicts rather than how a network trains, with a scroll-driven *story mode* through the hypotheses and an *explore mode* with the linked views of Figure 1.

(A) Canvas: probability, uncertainty (entropy) or boundary-only views, a far-field zoom-out, and a compare mode showing $|p_A - p_B|$. **(B) Small multiples:** a dataset-by-model overview with a linked crosshair; clicking loads a panel into the canvas. **(C) Controls:** dataset, model, seed, TabPFN version, rotation and noise toggles, and a training-size slider ($n = 10$–200) that animates boundaries as context grows. **(D) Metrics:** J, S, far-field confidence and ECE with seed ranges, plus, for the real datasets, the resemblance chart and before/after slope charts for rotation and noise. Values shown by color also appear as text; grids are downsampled to 100×100 and 8 bits for speed; Streamlit is the fallback.

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